Poisson Distribution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Conditions for a Poisson distribution**

State the key conditions required for a random variable to be modeled by a Poisson distribution.

Events occur independently of one another; events occur at a constant average rate over the given interval of time or space; two events cannot occur at exactly the same instant; the number of events in non-overlapping intervals is independent.

2 **[Foundational]** • **Target: Conditions for a Poisson distribution**

Explain why the number of texts a person receives in a randomly chosen hour might reasonably be modeled by a Poisson distribution, but the number of texts received right around a pre-arranged deadline might not.

Texts received during a typical hour occur independently and at a roughly constant average rate, satisfying the Poisson conditions. Texts clustered around a pre-arranged deadline (e.g. everyone replying just before a cutoff) violate the constant-rate and independence assumptions, since there is a systematic surge rather than random, uniform occurrence.

3 **[Foundational]** • **Target: Computing Poisson probabilities**

A shop receives an average of 3 customers per minute, modeled by $X \sim \text{Po}(3)$. Find $P(X = 5)$, to three decimal places.

$P(X = 5) = \frac{e^{-3}3^5}{5!} = \frac{(0.049787)(243)}{120} \approx 0.101.$

4 **[Foundational]** • **Target: Computing Poisson probabilities**

Using the same shop, find $P(X = 1)$, to three decimal places.

$P(X = 1) = 3e^{-3} \approx 3(0.049787) \approx 0.149.$

5 **[Foundational]** • **Target: Computing Poisson probabilities**

Using the same shop, find $P(X \leq 2)$.

$P(X = 0) = e^{-3} \approx 0.0498$. $P(X = 1) \approx 0.1494$. $P(X = 2) = \frac{9e^{-3}}{2} \approx 0.2240$. Sum: $P(X \leq 2) \approx 0.423$.

6 **[Foundational]** • **Target: Mean and variance of a Poisson distribution**

State the mean and variance of $X \sim \text{Po}(\lambda)$, and find the standard deviation when $\lambda = 9$.

$E(X) = \lambda$ and $\text{Var}(X) = \lambda$ — both equal λ, unlike most other distributions where mean and variance are independent parameters. When $\lambda = 9$: standard deviation $= \sqrt{9} = 3$.

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7 [Foundational] • Target: Mean and variance of a Poisson distribution

A random variable $X \sim \text{Po}(7)$. State $E(X)$, $\text{Var}(X)$, and the standard deviation, to three decimal places.

$E(X) = 7$, $\text{Var}(X) = 7$, standard deviation $= \sqrt{7} \approx 2.646$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sums of independent Poisson variables

Two independent Poisson processes have rates $\lambda_1 = 4$ and $\lambda_2 = 6$ per hour. State the distribution of the total number of events, and find its mean and variance.

The sum of independent Poisson variables is itself Poisson: total $\sim \text{Po}(4 + 6) = \text{Po}(10)$. Mean $= 10$; variance $= 10$.

9 [Intermediate] • Target: Sums of independent Poisson variables

Three independent Poisson processes have rates $\lambda_1 = 2$, $\lambda_2 = 4$, $\lambda_3 = 5$ per hour. Find the mean and variance of the total number of events across all three processes in one hour.

Total $\sim \text{Po}(2 + 4 + 5) = \text{Po}(11)$. Mean $= 11$; variance $= 11$.

10 [Intermediate] • Target: Poisson as an approximation to the binomial

Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the value of λ to use.

The approximation is appropriate when n is large and p is small (a common guideline: $n > 50$ and $np < 5$), so events are rare relative to the number of trials. Then X is approximately $\text{Po}(\lambda)$ with $\lambda = np$.

11 [Intermediate] • Target: Poisson as an approximation to the binomial

A factory produces 2000 items, each independently defective with probability 0.0025. Using the Poisson approximation, find the probability that exactly 2 items are defective, to three decimal places.

$\lambda = np = 2000(0.0025) = 5$. $P(X = 2) = \frac{e^{-5}5^2}{2!} = \frac{(0.006738)(25)}{2} \approx 0.084$.

12 [Intermediate] • Target: Poisson as an approximation to the binomial

A survey mails 5000 questionnaires, each independently containing a printing error with probability 0.0006. Using the Poisson approximation, find the probability that no questionnaires contain an error, to four decimal places.

$\lambda = np = 5000(0.0006) = 3$. $P(X = 0) = e^{-3} \approx 0.0498$.

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13 [Intermediate] • Target: Using cumulative Poisson probabilities

A call center receives calls at $\lambda = 6$ per 10-minute interval. Find the probability of receiving more than 3 calls, given $P(X = 0) \approx 0.00248$, $P(X = 1) \approx 0.01487$, $P(X = 2) \approx 0.04462$, $P(X = 3) \approx 0.08924$.

$P(X > 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)] = 1 - 0.15121 \approx 0.849$.

14 [Intermediate] • Target: Using cumulative Poisson probabilities

A website receives visitors at $\lambda = 4$ per minute. Find $P(1 \le X \le 3)$, given $P(X = 0) \approx 0.0183$, $P(X = 1) \approx 0.0733$, $P(X = 2) \approx 0.1465$, $P(X = 3) \approx 0.1954$.

$P(1 \le X \le 3) = P(X = 1) + P(X = 2) + P(X = 3) = 0.0733 + 0.1465 + 0.1954 \approx 0.415$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 13.01 Continuous Random Variables

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.0498$. Find λ, to the nearest whole number.

$e^{-\lambda} = 0.0498 \Rightarrow -\lambda = \ln(0.0498) \approx -3.00 \Rightarrow \lambda \approx 3$.

16 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 14.01 Sampling & Estimation

A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 2. Find λ, and then find $P(X = \lambda)$, to three decimal places.

Since variance $= \lambda$ and standard deviation $= \sqrt{\lambda} = 2$, $\lambda = 4$. $P(X = 4) = \frac{e^{-4}4^4}{4!} = \frac{(0.018316)(256)}{24} \approx 0.195$.

17 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 15.01 Hypothesis Testing

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 1) = 0.1494$, where λ is a whole number less than 5. Using $P(X = 1) = \lambda e^{-\lambda}$, determine λ by trial.

Testing $\lambda = 3$: $3e^{-3} = 3(0.049787) \approx 0.1494$ ✓, matching the given probability. So $\lambda = 3$.

18 [Challenging] • Target: Poisson processes over different intervals and conditional probability • Cross-Lesson: 2.01 Exponentials & Logarithms

Machine faults occur at an average rate of $\lambda = 2$ per day. Find the probability that exactly 2 faults occur today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.1353$ and $P(X = 2) \approx 0.2707$.

$P(X \ge 1) = 1 - P(X = 0) = 1 - 0.1353 = 0.8647$. Conditional probability:
$P(X = 2 \mid X \ge 1) = \frac{0.2707}{0.8647} \approx 0.313$.

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19 **[Challenging]** • **Target: Poisson processes over different intervals and conditional probability** • **Cross-Lesson: 3.01 Advanced Trigonometry**

A radioactive source emits particles at a rate of $\lambda = 1.2$ per second, modeled by a Poisson distribution. Find $P(X = 2)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

For a 3-second interval, $\lambda = 1.2 \times 3 = 3.6$. $P(X = 2) = \frac{e^{-3.6}(3.6)^2}{2!} = \frac{(0.027324)(12.96)}{2} \approx 0.177$.

20 **[Challenging]** • **Target: A well-designed word problem: quality control at a bottling plant** • **Cross-Lesson: 4.01 Differentiation**

This question has four parts. Faulty caps occur at random in bottling cases with an average rate of $\lambda = 1.5$ per case. (a) Find $P(X = 0)$ and $P(X = 1)$ for a randomly selected case, to four decimal places. (b) Find the probability that a case has at least 2 faulty caps. (c) The plant inspects 40 cases per day. Using the fact that a sum of independent Poisson variables is itself Poisson, state the distribution of the total number of faulty caps found across all 40 cases in a day, and find its mean. (d) Since this total has a large λ, it can be approximated by $N(\lambda, \lambda)$. Using a continuity correction and $\Phi(1.36) \approx 0.9131$, estimate the probability that more than 70 faulty caps are found in a day.

(a) $P(X = 0) = e^{-1.5} \approx 0.2231$. $P(X = 1) = 1.5e^{-1.5} \approx 0.3347$. (b) $P(X \geq 2) = 1 - 0.2231 - 0.3347 \approx 0.4422$. (c) Total $\sim \text{Po}(40 \times 1.5) = \text{Po}(60)$, with mean 60. (d) Using $N(60, 60)$ with $\sigma = \sqrt{60} \approx 7.746$, and a continuity correction for "more than 70" (i.e. ≥ 71): $z = \frac{70.5 - 60}{7.746} \approx 1.36$. $P(\text{total} > 70) \approx 1 - \Phi(1.36) \approx 1 - 0.9131 \approx 0.087$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Conditions for a Poisson distribution

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses $H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$ at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

a. $z = -2.0$**; reject** H_0 **because** $-2.0 < -1.645$ ✓

b. $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

c. $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

d. $z = -8.0$; reject H_0 because $-8.0 < -1.645$

1. Under the null hypothesis $H_0 : \mu = 25.0$, the sample mean $\bar{X}$ of a random sample of size $n = 16$ from $\text{N}(\mu, \sigma^2)$ has standard error: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{16}} = \frac{0.8}{4} = 0.2$ 2. The standardized test statistic z is: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{24.6 - 25.0}{0.2} = \frac{-0.4}{0.2} = -2.0$ 3. Since this is a lower-tail test at the 5% level with critical value $z_{\text{crit}} = -1.645$, the critical (rejec.

22 [Foundational] • Target: Computing Poisson probabilities

$X \sim Po(3)$ and $Y \sim Po(3)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 9

b. 0

c. **6** ✓

d. 7

Since X and Y are independent Poisson variables, $X + Y \sim Po(3 + 3) = Po(6)$.

23 [Foundational] • Target: Mean and variance of a Poisson distribution

A binomial random variable has $n = 200$, $p = \frac{1}{50}$, and is approximated by a Poisson distribution. Find the parameter λ of the approximating Poisson distribution.

a. 50

b. **4** ✓

c. 200

d. 5

$\lambda = np = 200 \times \frac{1}{50} = 4$.

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24 [Foundational] • Target: Sums of independent Poisson variables

The random variable X has a Poisson distribution with mean $\lambda > 0$. Given that $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$, find the value of λ.

a. 6 ✓

b. 36

c. 12

d. $\sqrt{6}$

For a Poisson distribution $X \sim \mathrm{Po}(\lambda)$, the probability mass function is given by: $\mathrm{P}(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}$ Substitute $r = 2$ and $r = 4$ into the given equation $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$: $3\left(\frac{e^{-\lambda}\lambda^2}{2!}\right) = \frac{e^{-\lambda}\lambda^4}{4!}$ Since $e^{-\lambda} > 0$ and $\lambda > 0$, divide both sides by $e^{-\lambda}\lambda^2$: $\frac{3}{2} = \frac{\lambda^2}{24}$ Multiply both sides by 24: $\lambda^2 = 36$ Since.

25 [Foundational] • Target: Poisson as an approximation to the binomial

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \leq k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999 .

a. 0.04

b. **0.08** ✓

c. 0.12

d. 0.16

Part (i): For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$. \\ We are given $n = 25$ and $E(X) = 2$. \\ So, $25p = 2$, which gives $p = \frac{2}{25} = 0.08$. \\ Part (ii): Now, $n = 50$ and $p = 0.08$. The number of defective bulbs follows $X \sim B(50, 0.08)$. \\ We need to find the smallest integer k such that $P(X \leq k) > 0.9$. \\ We can use a binomial cumulative distribution table or a calculator. \\ Let's calculate a few values: $P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$ Using.

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26 [Foundational] • Target: Using cumulative Poisson probabilities

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

a. **2.4** ✓

b. 3.6

c. 6.4

d. 7.2

For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$ and the variance is given by $Var(X) = np(1 - p)$. We are given that $n = 10$ and $E(X) = 4$. Using the formula for the mean, we have $10p = 4$. Solving for p, we get $p = \frac{4}{10} = 0.4$. Now we can find the variance using the formula $Var(X) = np(1 - p)$. Substituting the values of n and p, we get $Var(X) = 10 \times 0.4 \times (1 - 0.4)$. $Var(X) = 10 \times 0.4 \times 0.6$. $Var(X) = 4 \times 0.6$. $Var(X) = 2.4$.

27 [Foundational] • Target: Finding an unknown parameter

Given $X \sim Po(7)$, find $P(X \geq 1)$ in exact form.

a. $1 - e^{7}$

b. e^{-7}

c. $e^{7} - 1$

d. $1 - e^{-7}$ ✓

Applying Poisson Distribution principles, $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-7}$. Therefore, the verified correct option is $1 - e^{-7}$.

28 [Foundational] • Target: Poisson processes over different intervals and conditional probability

Given $X \sim Po(3)$, find an expression for $P(X = 5)$.

a. $\frac{3^6}{720}e^{-3}$

b. $\frac{3^4}{24}e^{-3}$

c. $\frac{3^5}{120}e^{3}$

d. $\frac{3^5}{120}e^{-3}$ ✓

$P(X = 5) = \frac{3^5}{120}e^{-3}$ by the Poisson formula $P(X = x) = \frac{\lambda^x}{x!}e^{-\lambda}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: quality control at a bottling plant

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

a. Yes — fixed, independent trials with constant success probability ✓

b. No — the number of trials is not fixed

c. No — trials are not independent / probability is not constant

d. Yes — but only if $n > 30$

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied.

30 [Intermediate] • Target: Conditions for a Poisson distribution

A random sample of 10 values of a variable x is taken. The results are summarised as follows: $\sum x = 50, \quad \sum x^2 = 322$ Calculate an unbiased estimate of the population variance.

a. 8 ✓

b. 7.2

c. 2.83

d. 32.2

To find an unbiased estimate of the population variance s^2, we use the formula: $s^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)$ Given $n = 10$, $\sum x = 50$, and $\sum x^2 = 322$: $\frac{(\sum x)^2}{n} = \frac{50^2}{10} = \frac{2500}{10} = 250$ Substitute these values into the formula: $s^2 = \frac{1}{10-1}(322 - 250) = \frac{1}{9} \times 72 = 8$ - The distractor 7.2 results from dividing by $n = 10$ instead of $n - 1 = 9$, which gives the biased sample variance rather than the unbiase.

31 [Intermediate] • Target: Computing Poisson probabilities

A random sample of $n = 50$ observations is taken from a population. The sample values are summarised as follows: $\sum x = 450, \quad \sum x^2 = 4295.$ Find an unbiased estimate of the population variance.

a. 5.0 ✓

b. 4.9

c. 2.24

d. 9.0

The unbiased estimate of the population variance, denoted s^2, is calculated using the formula: $s^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)$ Substitute the given values $n = 50$, $\sum x = 450$, and $\sum x^2 = 4295$: $s^2 = \frac{1}{50-1}\left(4295 - \frac{450^2}{50}\right)$ $s^2 = \frac{1}{49}\left(4295 - \frac{202500}{50}\right)$ $s^2 = \frac{1}{49}(4295 - 4050) = \frac{245}{49} = 5.0$ Why other choices fail: - 4.9 is the biased sample variance obtained by dividing by.

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32 [Intermediate] • Target: Mean and variance of a Poisson distribution

Formulate the null and alternative hypotheses for the following scenario: "A factory claims the mean weight of a product is 250g. An inspector wants to test if it has changed (either direction)."

a. $H_0 : \mu = 500,\ H_1 : \mu < 500$

b. $H_0 : \mu = 3,\ H_1 : \mu > 3$

c. $H_0 : \mu = 10,\ H_1 : \mu < 10$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$ ✓

Applying Hypothesis Testing principles, $H_0 : \mu = 250,\ H_1 : \mu e 250$. Therefore, the verified correct option is $H_0 : \mu = 250,\ H_1 : \mu e 250$.

33 [Intermediate] • Target: Sums of independent Poisson variables

Formulate the null and alternative hypotheses for the following scenario: "A manufacturer claims the mean lifetime of a battery is 500 hours. A researcher suspects it is less than this."

a. $H_0 : \mu = 500,\ H_1 : \mu < 500$ ✓

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 3,\ H_1 : \mu > 3$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$

Applying Hypothesis Testing principles, $H_0 : \mu = 500,\ H_1 : \mu < 500$. Therefore, the verified correct option is $H_0 : \mu = 500,\ H_1 : \mu < 500$.

34 [Intermediate] • Target: Poisson as an approximation to the binomial

Formulate the null and alternative hypotheses for the following scenario: "A school claims the mean test score is 70. A teacher believes it is actually higher."

a. $H_0 : \mu = 10,\ H_1 : \mu < 10$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$ ✓

c. $H_0 : \mu = 500,\ H_1 : \mu < 500$

d. $H_0 : \mu = 3,\ H_1 : \mu > 3$

Applying Hypothesis Testing principles, $H_0 : \mu = 70,\ H_1 : \mu > 70$. Therefore, the verified correct option is $H_0 : \mu = 70,\ H_1 : \mu > 70$.

35 [Intermediate] • Target: Using cumulative Poisson probabilities

Using the trapezium rule with 3 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 1, y_1 = 1, y_2 = 2, y_3 = 10$.

a. $\frac{17}{2}$ ✓

b. 7

c. $\frac{103}{12}$

d. $\frac{19}{2}$

$\int y\,dx \approx \frac{h}{2}[y_0 + y_3 + 2(y_1 + \cdots + y_2)] = \frac{1}{2} \times 17 = \frac{17}{2}$.

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36 [Intermediate] • Target: Finding an unknown parameter

Find $\int x(x^2+1)^2\,dx$ using substitution.

a. $\frac{x^2(x^2+1)^2}{2}+C$

b. $\frac{x^6}{6}+\frac{x^4}{2}+\frac{x^2}{2}+C$ ✓

c. $\frac{2(x^2+1)^3}{3}+C$

d. $\frac{(x^2+1)^3}{3}+C$

Let $u=x^2+1$, $du=2x\,dx$: $\int x(x^2+1)^2dx=\frac{x^6}{6}+\frac{x^4}{2}+\frac{x^2}{2}+C$.

37 [Intermediate] • Target: Poisson processes over different intervals and conditional probability

A continuous random variable X has PDF $f(x)=\frac{2}{2^2}x$ for $0\le x\le 2$. Find the median of X in exact form.

a. $\frac{2\sqrt{2}}{2}$ ✓

b. $\frac{2\sqrt{2}}{4}$

c. $\frac{2}{2}$

d. $2\sqrt{2}$

$F(m)=(\frac{m}{2})^2=\frac{1}{2}\Rightarrow m=\frac{2}{\sqrt{2}}=\frac{2\sqrt{2}}{2}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: quality control at a bottling plant • Cross-Lesson: 2.01 Exponentials & Logarithms

A sample of size $n=9$ is drawn from a population with mean $\mu=42$ and $\sigma=8$. Given $\bar{x}=\frac{118}{3}$, find $Z=\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$.

a. 0

b. 1

c. **-1** ✓

d. -3

$\sigma/\sqrt{n}=8/3=\frac{8}{3}$. $\bar{x}-\mu=-\frac{8}{3}$. $Z=-\frac{8}{3}\div\frac{8}{3}=-1$.

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39 [Challenging] • Target: Conditions for a Poisson distribution • Cross-Lesson: 3.01 Advanced Trigonometry

A continuous random variable X has PDF $f(x) = kx^1$ for $0 \le x \le 4$ (and 0 otherwise). Find k.

a. $\frac{5}{24}$

b. $\frac{1}{4}$

c. $\frac{1}{16}$

d. $\frac{1}{8}$ ✓

$\int_0^4 kx^1 dx = k\frac{4^2}{2} = 1 \Rightarrow k = \frac{2}{16} = \frac{1}{8}$.

40 [Challenging] • Target: Computing Poisson probabilities • Cross-Lesson: 4.01 Differentiation

A continuous random variable X has PDF $f(x) = \frac{2}{4^2}x^1$ for $0 \le x \le 4$. Find $\text{Var}(X)$.

a. $\frac{17}{9}$

b. $\frac{8}{9}$ ✓

c. $\frac{8}{3}$

d. 8

$E(X) = \frac{8}{3}$, $E(X^2) = 8$. $\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{8}{9}$.

41 [Challenging] • Target: Mean and variance of a Poisson distribution • Cross-Lesson: 5.01 Integration

A sample of size 6 has $\sum x = 282$ and $\sum x^2 = 13312$. Find the unbiased estimate of the population variance.

a. $\frac{6656}{3}$

b. $\frac{58}{5}$ ✓

c. $\frac{63}{5}$

d. $\frac{29}{3}$

$s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1} = \frac{13312 - 282^2/6}{5} = \frac{58}{5}$.

42 [Challenging] • Target: Sums of independent Poisson variables • Cross-Lesson: 6.01 Sequences & Series

$X_1, \ldots, X_{14}$ are independent and identically distributed with $\text{Var}(X) = 3$. Find $\text{Var}(X_1 + \cdots + X_{14})$.

a. **42** ✓

b. 43

c. 3

d. 39

$\text{Var}\left(\sum X_i\right) = n\,\text{Var}(X) = 14 \times 3 = 42$.

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43 [Challenging] • Target: Poisson as an approximation to the binomial • Cross-Lesson: 7.01 Vectors

A hypothesis test is carried out at the 10% significance level. Find the probability of a Type I error.

a. 90\%

b. **10\%** ✓

c. 50\%

d. 0\%

By definition, $P(\text{Type I error}) = \alpha = 10\%$.

44 [Challenging] • Target: Using cumulative Poisson probabilities • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A sample of size 6 gives values: 5, 12, 18, 11, 17, 18. Find the sample mean $\bar{x}$.

a. 18

b. $\frac{29}{2}$

c. $\frac{27}{2}$ ✓

d. $\frac{81}{7}$

$\bar{x} = \frac{\sum x}{n} = \frac{81}{6} = \frac{27}{2}$.

45 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 9.01 Complex Numbers

A confidence interval for the mean uses $z = 1$, $\sigma = 8$, $n = 9$. Find the total width of the confidence interval, $2E$.

a. $\frac{19}{3}$

b. $\frac{16}{3}$ ✓

c. 8

d. $\frac{8}{3}$

$E = \frac{8}{3}$, so width $= 2E = \frac{16}{3}$.

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